

Operations Research: theory and applications to networking

Covering

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ICT4SS-CE

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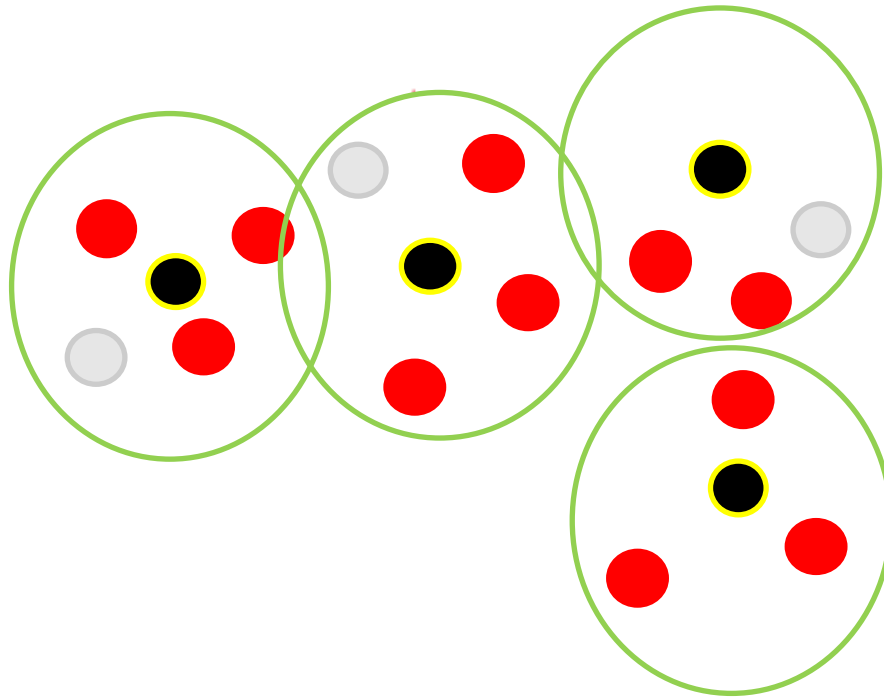
Given a set of demand points distributed in a geometric space (for example, cities on a map), and a set of candidate locations where you could place facilities (such as warehouses, antennas, or service centers).

Each facility can serve all demand points that lie within a fixed distance (a coverage radius).

The goal is to choose a subset of these candidate locations so that every demand point is covered by at least one selected facility, while minimizing the total cost (or simply the number) of facilities used.

An example

Formulate the problem



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Both N demand points and M facility locations are chosen independently and uniformly at random in $[0,1]^2$.

Costs associated to location facilities are i.i.d and uniform in $[1,10]$;

Choose:

$N=60, M=25 R=0.2$

$N=50, M=35 R=0.15$.

and generate 3/4 instances.

Formulation extension

Consider now a **partial covering problem**, where only a fraction α of demand points must be covered.

Adapt your formulation to handle this new case.